

Group Project Specification (10% in Overall Marks)

Title: Domain-specific Chatbot Development

Submission Deadlines: 18 Nov 2026, 23:59 (Thu Class)

19 Nov 2026, 23:59 (Fri Class)

A. Requirements

In this project, you will work in groups to design, implement, and evaluate an AI-powered chatbot tailored for a specific application domain (e.g., healthcare, education, customer service, finance, travel, etc.). The requirements for your chatbot are outlined below:

Basic Features (3 marks out of 10 marks in Final Demo)

Your chatbot should be capable of understanding user queries, providing relevant information, and performing essential tasks within the selected domain. Specifically:

- The chatbot must be able to communicate effectively within the defined domain of application.
- It should handle typical user queries and provide relevant responses in a chosen language (e.g. English or Chinese).

Additional Features (7 out of 10 marks in Final Demo)

Beyond the basic requirements, you are expected to incorporate additional features to enhance your chatbot. You have the flexibility to select any features you prefer, such as (but not limited to):

- Voice recognition and synthesis
- 2D/3D avatar integration
- Fine-tuned large language model (LLM)
- User-friendly interface

Innovation and technical complexity in your chosen features will be considered in the evaluation and may positively impact your marks.

B. Group Formation

You are required to form groups of **2-4 students**. The workload should be equally shared among all group members, and as such, each student in the group will receive the same project mark.

C. Project Development

Final Demo and Presentation [10 marks]

Presentation

Clearly state all the features you have implemented, including both the basic and additional features.

Additionally, provide a detailed description of how the additional features were implemented or integrated into your chatbot. Highlight any technical challenges you encountered and how you addressed them. Explain their capabilities as well as any limitations or challenges encountered during development.

Live Demonstration

You are required to conduct a live demonstration of your developed chatbot. Please design a scenario or specific use case that effectively showcases the capabilities of your chatbot within the chosen domain.

For example, you may interact with your chatbot using domain-specific queries to demonstrate its ability to provide accurate responses based on your knowledge base or vector database. If you have implemented features such as speech recognition, consider including a verbal conversation to highlight this functionality.

If applicable, please also summarize any evaluation results and user feedback during your demonstration.

Grading Criteria:

Your final demo and presentation will be evaluated based on the following aspects:

- Technical contributions
 - complexity and difficulty of implementation, effectiveness and performance of the developed features
- User interface design
 - visual appeal, usability, intuitiveness, integration of features such as speech recognition
- Creativity and innovation in features
 - originality of features, how well they enhance user experience or address the specific needs of the chosen domain
- Quality of the final presentation and demonstration
 - clarity, organization, engagement, and effectiveness in showcasing the chatbot's capabilities

Important Dates and Submission of Assessments:

- **Group Forming (25 Sep 2026, Friday):**

Submit your group's member list via the online form below:

Thu Class : <https://forms.cloud.microsoft/r/kmk8egxVE9>

Fri Class : <https://forms.cloud.microsoft/r/2n0cQJ6dcz>

Please input **BOTH the student ID and English names** of your members. Each group needs ONLY to submit **ONCE!**

- **Demo and Presentation Submission**

(18 Nov 2026, Wednesday, 23:59 for Thu Class)

(19 Nov 2026, Thursday, 23:59 for Fri Class):

Submit the source file package (in zip file) and presentation slide (in pptx) via Canvas.

For your source code, **Include a readme file** to include all the **student IDs and names** of all group members in it. Then describe the additional feature you had implemented and the way you run your source code. If you are using an online platform to develop your chatbot, submit your prompt, related knowledge base files in your source file package.

Each group ONLY needs to submit one copy of the item **by one group member** via Canvas.

- **Live Demo and Presentation**

(19 or 26 Nov 2026, Thursday for Thu Class)

(20 or 27 Nov 2026, Friday for Fri Class):

Present and demonstrate your finished chatbot in **English within 7 minutes (tentative) and 3 minus for Q&A**. Arrange for at **least one group member** to present during the lecture session. The exact time slot will be arranged randomly a week before the presentation.